Sahan S Liyanage

Physics Graduate | Seeking Graduate Study Opportunities

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Physics graduate from the University of Colombo seeking graduate study opportunities in physics. Experience in solar-radio research, optical spectroscopy, scientific software development, university teaching, and research mentoring.

EDUCATION

Bachelor of Science Honours in Physics - Second Class (Upper Division)

Jan 2020 - May 2024

University of Colombo, Sri Lanka

- Cumulative GPA: 3.57/4.00 (126 credit hours).
- Relevant coursework included quantum mechanics, classical mechanics, electromagnetic fields, statistical physics, advanced optics, solid-state physics, nuclear and particle physics, computational physics, electronics, data acquisition, signal processing, and advanced physics laboratories.

Secondary Education

2015 - 2017

Central College, Anuradhapura, Sri Lanka

- GCE Advanced Level (2018), Physical Sciences: Combined Mathematics - B, Physics - B, Chemistry - B.

PROFESSIONAL EXPERIENCE

Research Assistant - Astronomy and Space Applications Division

May 2026 - Present

Arthur C. Clarke Institute for Modern Technologies, Katubedda, Moratuwa, Sri Lanka

- Conduct solar-radio observations with the e-CALLISTO antenna system, including calibration, performance checks, and hardware maintenance.
- Carry out H-alpha solar observations and record, process, and interpret the resulting images.
- Perform astronomical spectroscopy with optical telescopes and reduce and interpret the resulting spectra.
- Develop software and automated pipelines for observational data acquisition, processing, and archiving.
- Support public workshops and observation sessions in space physics and astronomy.

RESEARCH EXPERIENCE

Undergraduate Research Project

May 2023 - Apr 2024

Department of Physics, University of Colombo

Determination of the Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts

- Analyzed 25 metric Type II solar-radio-burst datasets from the e-CALLISTO network.
- Developed a Python workflow for FITS reading, background-noise reduction, burst isolation, and dynamic-spectrum analysis.
- Derived shock-speed and source-height expressions using plasma-oscillation theory and the Newkirk coronal-density model.
- Estimated shock speeds of 343-1032 km/s and source heights of 1.317-1.724 solar radii.
- Supervisors: Dr. Janaka Adassuriya and Prof. K. P. S. C. Jayaratne.

Raman Spectroscopy Project

Nov 2022 - Mar 2023

Department of Physics, University of Colombo

Raman Spectroscopic Analysis of Common Minerals in Sri Lanka

- Analyzed calcite, dolomite, apatite, quartz, feldspar, graphite, and mica using a 532 nm Raman system.
- Prepared samples and processed spectra in Python using calibration, baseline correction, smoothing, and plotting.
- Supervisor: Prof. H. H. E. Jayaweera.

Undergraduate Research Mentor

May 2025 - Apr 2026

Final-year project mentoring

Estimating Coronal Magnetic Field Strength from Band Splitting in Type II Solar Radio Bursts

PROJECTS AND SCIENTIFIC SOFTWARE

Space Weather Monitoring Dashboard
2026 - Present

Full-stack solar and geospace monitoring application

- Combines e-CALLISTO spectra, GOES flux, SDO and SOHO imagery, Kp and Dst indices, and solar-wind and interplanetary magnetic-field data with alerts, a combined timeline, and archive search.
- ResNet-18 models achieved a test F1 score of 0.932 for burst detection and 0.966 validation accuracy with macro F1 0.889 for three-class burst typing.
- github.com/SaanDev/Solar_Monitoring_Dashboard

ecallistolib
2025 - Present

MIT-licensed Python library, version 1.3.1

- Downloads, reads, processes, combines, and plots e-CALLISTO FITS dynamic spectra with metadata-preserving parsing, RFI mitigation, background subtraction, and publication-quality output.
- Supports Python 3.10 through 3.14, a command-line interface, optional accelerated dependencies, and a pytest test suite.
- github.com/SaanDev/ecallistolib

e-CALLISTO FITS Analyzer
2024 - Present

Peer-reviewed, open-source Python desktop application

- Cross-platform framework for retrieving, cleaning, combining, visualizing, and analyzing e-CALLISTO dynamic spectra at event level.
- Uses Python, PySide6, NumPy, SciPy, Astropy, SunPy, Matplotlib, PyQtGraph, netCDF4, OpenPyXL, Requests, and BeautifulSoup.
- github.com/SaanDev/e-Callisto_FITS_Analyzer
- ecallistoanalyzer.org

Spectral Analysis of Fluids to Determine the Effects of Temperature
2021

General Physics Laboratory II, University of Colombo

- Used a home-made spectrometer and image-based absorbance measurements to estimate liquid temperature.
- Distilled water at 462 nm produced the best response: T = 73A + 230, with model errors of 4-12%.

PUBLICATIONS

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., Monstein, C., & Manoharan, P. K. (2026). e-CALLISTO FITS Analyzer: A software framework for CALLISTO solar radio data. *RAS Techniques and Instruments*, rzag056. doi:10.1093/rasti/rzag056

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., & Monstein, C. (2025). Determination of speed and source height of coronal shock waves using Type II solar radio bursts. *Proceedings of the Technical Sessions*, 41, 59-67. Institute of Physics, Sri Lanka. doi:10.48550/arXiv.2504.00622

SCIENTIFIC PRESENTATIONS

Determination of Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts
22 Mar 2025

41st Technical Sessions, Institute of Physics, Sri Lanka - University of Colombo

- On-site oral presentation of undergraduate research findings.

Solar Radio Bursts and Space Weather
20 Nov 2024

Physics Society, University of Colombo

- On-site oral presentation for physics undergraduates.

Introduction to Solar Physics
18 Jul 2024

Astronomical Society, Rajarata University of Sri Lanka

- Online introductory lecture for university students.

TEACHING EXPERIENCE

Visiting Lecturer
Oct 2026 - Present

Department of Physics, University of Colombo

- Visiting lecturer for the Certificate in Astronomy course.

Teaching Assistant
Aug 2025 - Mar 2026

University of Colombo School of Computing

- Probability and Statistics for first-year Computer Science majors.
- Digital Signal Processing for second-year Computer Science majors.
- Cryptography and Information Security for second-year Computer Science majors.

Teaching Assistant
May 2024 - May 2025

Department of Physics, University of Colombo

- Instructor in charge of Computational Electrodynamics and Advanced Physics Laboratory I.
- Instructor for Analogue and Digital Electronic Principles Laboratory.
- Instructor for General Physics Laboratory I and II.

ACADEMIC ENGAGEMENTS

Instructor - COSPAR Capacity Building Workshop 2026
31 Aug - 11 Sep 2026

National Research Institute of Astronomy and Geophysics, Cairo, Egypt

- Taught solar-radio data analysis using Python at the workshop on Earth-Affecting Solar Transients.

Participant - COSPAR Capacity Building Workshop 2024
19-30 Aug 2024

Samarkand State University, Samarkand, Uzbekistan

- Practical training on coronal and interplanetary shocks using SOHO, SDO, STEREO, ground-based observations, SolarSoft, SunPy, AstroPy, and Python.

Designing Physics Laboratory Experiments
General Physics Laboratory II, Department of Physics, University of Colombo

- Designed the first-year optical-spectrometer laboratory sheet, including theory, pre-lab questions, prisms, refractive index, diffraction gratings, and atomic emission spectra.

SKILLS AND TECHNIQUES

Programming and scientific computing: Python, C/C++, MATLAB, SunPy, AstroPy, NumPy, SciPy, Matplotlib, FITS handling, signal processing, and Qt GUI development.

Software and tools: OriginPro, SolidWorks, Microsoft Office, Adobe Photoshop, LaTeX, Arduino, Linux, and web development.

Laboratory and instrumentation: Raman and UV-Vis spectroscopy, astronomical spectroscopy, e-CALLISTO radio spectrometry, analogue and digital electronics, data acquisition, signal processing, and microcontroller programming.

Management and leadership: Communication, project planning and execution, teaching, mentoring, and team coordination.

AWARDS AND HONOURS

Dean's List
Academic Year 2019/2020

University of Colombo

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Vice President
2022 - 2023

Astronomical Society, University of Colombo

Coordinator
2020 - 2022

Astronomical Society, University of Colombo

Music Director
2022 - 2023

Music Society, University of Colombo

Marketing Director
2020 - 2021

Leo Club, University of Colombo

VOLUNTEERING EXPERIENCE

Science Bus - Phases I, II, and III
2021 - 2023

Physics Society, University of Colombo

- Contributed to an SPIE-funded travelling science project for school students in rural Sri Lanka.

Star Quest - All-Island Astronomical Quiz

2022 - 2025

Astronomical Society, University of Colombo

- Helped create questions, coordinate participants, run the event, and grade entries from school students across Sri Lanka.

Naada Nu 23 - Annual Musical Concert

2023

Music Society, University of Colombo

- Served as music director, contributed to event organization, and performed on stage.